

US-003 — Create Google Spreadsheet

1. Objective

The objective of US-003 was to create the initial Google Spreadsheet data store for the **Personal Finance & Money Flow Tracker**.

The spreadsheet provides the foundational data structure that will be used by the application in later user stories.

The implementation was intentionally limited to creating the spreadsheet structure, defining the required columns, validating the column order, and preparing initial/sample data.

Advanced data validation and dropdown rules were intentionally deferred until the master data is finalized.

2. Spreadsheet Created

A Google Spreadsheet named:

PersonalFinanceTracker

was created.

The spreadsheet contains four required worksheets:

```
PersonalFinanceTracker
|
├─ Accounts
├─ Categories
├─ Transactions
└─ Configuration
```

The worksheet order was also verified to match the expected structure.

3. Accounts Sheet

The `Accounts` sheet stores the user's financial accounts and their opening balances.

The following columns were created in the exact order defined in the TDD:

Column	Column Name	Description
A	Account_ID	Unique account identifier
B	Account_Name	Account display name
C	Account_Type	Account type: BANK / CREDIT_CARD / GIFT_CARD
D	Opening_Balance	Initial balance/value

Final structure

```

Accounts
├─ Account_ID
├─ Account_Name
├─ Account_Type
└─ Opening_Balance

```

The column ordering was verified against the TDD.

Initial/sample account data was added to establish the base master-data structure.

4. Categories Sheet

The `Categories` sheet stores the predefined transaction categories used by the application.

The following columns were created:

Column	Column Name	Description
A	Category_ID	Unique category identifier
B	Category_Name	Category display name

Final structure

```

Categories
├─ Category_ID
└─ Category_Name

```

The column ordering was verified against the TDD.

Initial/sample category master data was added.

5. Transactions Sheet

The `Transactions` sheet is the primary operational table used to record financial transactions.

The complete transaction schema was created according to the TDD.

Column	Column Name	Description
A	<code>Transaction_ID</code>	Unique transaction identifier
B	<code>Transaction_Date</code>	Business transaction date
C	<code>Transaction_Type</code>	EXPENSE / INCOME / TRANSFER
D	<code>Amount</code>	Transaction amount
E	<code>Category_ID</code>	Category reference
F	<code>Payment_Method</code>	BANK / CREDIT_CARD / GIFT_CARD
G	<code>Paid_From_Account_ID</code>	Expense source account
H	<code>Received_Into_Account_ID</code>	Income destination account
I	<code>From_Account_ID</code>	Transfer source
J	<code>To_Account_ID</code>	Transfer destination
K	<code>Notes</code>	Optional notes
L	<code>Created_Date</code>	Technical record creation timestamp
M	<code>Updated_Date</code>	Technical record update timestamp
N	<code>Status</code>	ACTIVE / DELETED

Final structure

```
Transactions
├─ Transaction_ID
├─ Transaction_Date
├─ Transaction_Type
├─ Amount
├─ Category_ID
├─ Payment_Method
├─ Paid_From_Account_ID
├─ Received_Into_Account_ID
├─ From_Account_ID
├─ To_Account_ID
├─ Notes
├─ Created_Date
├─ Updated_Date
└─ Status
```

The column sequence was explicitly checked to ensure that it matches the TDD.

This is important because later application logic will depend on this defined transaction structure.

6. Configuration Sheet

The `Configuration` sheet was created to store application-level configuration values.

The following columns were added:

Column	Column Name	Description
A	<code>Config_Key</code>	Configuration key
B	<code>Config_Value</code>	Configuration value
C	<code>Description</code>	Configuration description

Final structure

```
Configuration
├─ Config_Key
├─ Config_Value
└─ Description
```

The column order was verified against the TDD.

7. Column Ordering Verification

After creating all four worksheets, each sheet was reviewed against the TDD.

The following were verified:

Accounts

```
A → Account_ID
B → Account_Name
C → Account_Type
D → Opening_Balance
```

Categories

A → Category_ID
B → Category_Name

Transactions

A → Transaction_ID
B → Transaction_Date
C → Transaction_Type
D → Amount
E → Category_ID
F → Payment_Method
G → Paid_From_Account_ID
H → Received_Into_Account_ID
I → From_Account_ID
J → To_Account_ID
K → Notes
L → Created_Date
M → Updated_Date
N → Status

Configuration

A → Config_Key
B → Config_Value
C → Description

The final spreadsheet structure therefore matches the expected TDD schema.

8. Initial / Sample Master Data

Initial/sample data was added to establish the master-data structure.

The purpose of this data at this stage is to verify that:

1. The columns accept the expected values.
2. The relationships between master data and transactions can be represented.
3. The spreadsheet structure is ready for subsequent development.
4. Later user stories can build upon the established schema.

For example, transaction records can later reference:

```
Transactions.Category_ID
      ↓
Categories.Category_ID
```

and:

```
Transactions.Paid_From_Account_ID
      ↓
Accounts.Account_ID
```

This establishes the basic reference relationship between the operational transaction table and the master tables.

9. Data Validation — Intentionally Deferred

Data validation was **not implemented as part of US-003**.

This was an intentional implementation decision.

At this stage, the master-data tables are still being developed and finalized through subsequent user stories.

Therefore, dropdowns and validation rules such as:

```
Account_Type
      ↓
BANK
CREDIT_CARD
GIFT_CARD
```

or:

```
Transaction_Type
      ↓
EXPENSE
INCOME
TRANSFER
```

or account/category reference dropdowns were not added yet.

Reason

Applying validation before the master data is finalized could result in:

- Temporary validation ranges
- Rework when master data changes
- Incorrect references
- Duplicate configuration
- Additional maintenance

Therefore, validation will be implemented in the appropriate later user story after the master data structure and values are finalized.

The planned flow is:

```
US-003
Spreadsheet Structure
    ↓
Master Data Completion
    ↓
Accounts / Categories finalized
    ↓
Validation Rules
    ↓
Dropdowns & Reference Validation
    ↓
Application Integration
```

10. Basic Spreadsheet Preparation

The spreadsheet was also prepared for easier future use.

The initial preparation included:

- Creating the required worksheets.
- Maintaining the required worksheet order.
- Adding the required headers.
- Verifying column ordering.
- Adding initial/sample data.
- Preparing the sheets for future master-data and transaction development.

Data validation and advanced automation were intentionally excluded from this stage.

11. Final Spreadsheet Structure

The completed US-003 structure is:

```

PersonalFinanceTracker
├── Accounts
│   ├── Account_ID
│   ├── Account_Name
│   ├── Account_Type
│   └── Opening_Balance
├── Categories
│   ├── Category_ID
│   └── Category_Name
├── Transactions
│   ├── Transaction_ID
│   ├── Transaction_Date
│   ├── Transaction_Type
│   ├── Amount
│   ├── Category_ID
│   ├── Payment_Method
│   ├── Paid_From_Account_ID
│   ├── Received_Into_Account_ID
│   ├── From_Account_ID
│   ├── To_Account_ID
│   ├── Notes
│   ├── Created_Date
│   ├── Updated_Date
│   └── Status
└── Configuration
    ├── Config_Key
    ├── Config_Value
    └── Description

```

12. US-003 Completion Status

US-003 is considered completed because the required spreadsheet structure has been established and verified.

Completed

- [x] Created `PersonalFinanceTracker`
- [x] Created `Accounts` sheet
- [x] Created `Categories` sheet
- [x] Created `Transactions` sheet
- [x] Created `Configuration` sheet
- [x] Added required headers
- [x] Verified column ordering against TDD

- [x] Added initial/sample master data
- [x] Established the base data structure
- [x] Prepared the spreadsheet for subsequent development

Deferred to Later US

- [] Data validation
 - [] Dropdown lists
 - [] Account reference validation
 - [] Category reference validation
 - [] Transaction-type validation
 - [] Final master-data population
 - [] Advanced spreadsheet formulas
 - [] Application/API integration
-

13. Reference Spreadsheet

The implemented spreadsheet is:

PersonalFinanceTracker

Google Sheets: https://docs.google.com/spreadsheets/d/1KWPCabyEjVZThiBRNrXy_3jr18wh6iG2dV3v4ddWjrI/edit

This spreadsheet represents the base data-storage structure for the subsequent implementation phases of the Personal Finance & Money Flow Tracker.